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THERMAL RUNAWAY PROTECTION SYSTEM FOR A BATTERY PACK

Abstract

According to several aspects, a thermal runaway ducting system for a battery pack having one or more thermal runaway protection (TRP) trays is provided. The thermal runaway ducting system may include one or more flexible hoses in fluid communication with the one or more TRP trays. The thermal runaway ducting system further may include a sensor manifold in fluid communication with the one or more flexible hoses and a battery pack exit vent.

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Background/Summary

INTRODUCTION

[0001] The present disclosure relates to systems for thermal runaway protection (TRP) for battery packs, and more particularly, to thermal runaway ducting systems for battery packs having one or more thermal runaway protection (TRP) trays.

[0002] To increase performance, manufacturability, and reliability of vehicle battery packs, vehicle battery packs may be equipped with thermal runaway protection (TRP) systems which are designed to mitigate the buildup of fluid (e.g., gas) released from battery cells during a thermal runaway event. Current TRP systems may include one or more ducts or passageways to direct fluid towards an exit vent of the battery pack. TRP systems further may include insulating material for the purpose of preventing thermal runaway of one battery cell from propagating to adjacent battery cells. In some examples, the insulating material includes an expanding potting compound. However, the one or more ducts or passageways of current TRP systems may not be fluid-tight, allowing the expanding potting compound to intrude into the one or more ducts or passageways, reducing a volume of the passageways and thus impeding fluid flow.

[0003] Thus, while TRP systems achieve their intended purpose, there is a need for a new and improved system for thermal runaway protection (TRP) for battery packs.

SUMMARY

[0004] According to several aspects, a thermal runaway ducting system for a battery pack having one or more thermal runaway protection (TRP) trays is provided. The thermal runaway ducting system may include one or more flexible hoses in fluid communication with the one or more TRP trays. The thermal runaway ducting system further may include a sensor manifold in fluid communication with the one or more flexible hoses and a battery pack exit vent.

[0005] In another aspect of the present disclosure, each of the one or more flexible hoses further includes a first quick connect fitting, a second quick connect fitting, and a flexible tube in fluid communication with the first quick connect fitting and the second quick connect fitting. The flexible tube provides fluid communication between the first quick connect fitting and the second quick connect fitting.

[0006] In another aspect of the present disclosure, the one or more flexible hoses further includes a first flexible hose. The first quick connect fitting of the first flexible hose is in fluid communication with a first TRP tray. The second quick connect fitting of the first flexible hose is in fluid communication with the sensor manifold. The one or more flexible hoses further includes a second flexible hose. The first quick connect fitting of the second flexible hose is in fluid communication with a second TRP tray. The second quick connect fitting of the second flexible hose is in fluid communication with the sensor manifold.

[0007] In another aspect of the present disclosure, the sensor manifold further includes a first inlet port in fluid communication with the second quick connect fitting of the first flexible hose. The sensor manifold further includes a second inlet port in fluid communication with the second quick connect fitting of the second flexible hose.

[0008] In another aspect of the present disclosure, the one or more flexible hoses further includes a first flexible hose. The first quick connect fitting of the first flexible hose is in fluid communication with a first TRP tray. The second quick connect fitting of the first flexible hose is in fluid communication with a second TRP tray. The one or more flexible hoses further includes a second flexible hose. The first quick connect fitting of the second flexible hose is in fluid communication with the second TRP tray. The second quick connect fitting of the second flexible hose is in fluid communication with the sensor manifold.

[0009] In another aspect of the present disclosure, the sensor manifold further includes a first inlet port in fluid communication with the second quick connect fitting of the second flexible hose.

[0010] In another aspect of the present disclosure, the one or more flexible hoses further includes further includes a tee-joint hose in fluid communication with a first TRP tray and a second TRP

tray. The one or more flexible hoses further includes a first flexible hose. The first quick connect fitting of the first flexible hose is in fluid communication with the tee-joint hose. The second quick connect fitting of the first flexible hose is in fluid communication with the sensor manifold.

[0011] In another aspect of the present disclosure, the sensor manifold further includes a first inlet port in fluid communication with the second quick connect fitting of the first flexible hose.

[0012] In another aspect of the present disclosure, the sensor manifold further includes one or more inlet ports in fluid communication with one of the one or more flexible hoses. The sensor manifold further includes a sensor port configured to allow measurement of a fluid within the sensor manifold using a sensor. The sensor manifold further includes an outlet port in fluid communication with the battery pack exit vent.

[0013] In another aspect of the present disclosure, the thermal runaway ducting system further includes a battery pack casting. The battery pack exit vent is integral with the battery pack casting. The outlet port of the sensor manifold is in fluid communication with the battery pack exit vent using a press-in-place seal.

[0014] In another aspect of the present disclosure, the thermal runaway ducting system further includes a battery pack casting. The battery pack exit vent is integral with the battery pack casting. The sensor manifold is integral with the battery pack casting.

[0015] According to several aspects, a thermal runaway ducting system for a battery pack is provided. The thermal runaway ducting system may include one or more thermal runaway protection (TRP) trays. The thermal runaway ducting system further may include one or more flexible hoses in fluid communication with the one or more TRP trays using quick connect fittings. The thermal runaway ducting system further may include a sensor manifold in fluid communication with the one or more flexible hoses and a battery pack exit vent using quick connect fittings.

[0016] In another aspect of the present disclosure, each of the one or more TRP trays further includes an outlet spigot. The outlet spigot has a first portion disposed within one of the one or more TRP trays and a second portion disposed outside of one of the one or more TRP trays. The second portion is in fluid communication with the one or more flexible hoses.

[0017] In another aspect of the present disclosure, the thermal runaway ducting system of claim **13**, the first portion of the outlet spigot further includes a solid particulate filter for preventing solid particles from entering the outlet spigot.

[0018] In another aspect of the present disclosure, the second portion of the outlet spigot further includes a quick connect fitting configured to engage with one of the one or more flexible hoses.

[0019] In another aspect of the present disclosure, the quick connect fitting is substantially perpendicular to one of the one or more TRP trays.

[0020] In another aspect of the present disclosure, each of the one or more TRP trays further includes a tray portion having a convex surface. Each of the one or more TRP trays further includes a lid portion having an interior lid surface. The lid portion is affixed to the tray portion such that a fluid collection cavity is formed between the interior lid surface of the lid portion and the convex surface of the tray portion. Each of the one or more TRP trays further includes one or more filtering fin members disposed on the interior lid surface of the lid portion. The one or more filtering fin members are configured to prevent solid particles from entering the outlet spigot.

[0021] According to several aspects, a thermal runaway ducting system for a battery pack is provided. The thermal runaway ducting system may include one or more thermal runaway protection (TRP) trays. Each of the one or more TRP trays has an outlet spigot. The outlet spigot has a first portion disposed within one of the one or more TRP trays and a second portion disposed outside of one of the one or more TRP trays. The second portion of the outlet spigot has a quick connect fitting. The thermal runaway ducting system further may include one or more flexible hoses in fluid communication with the outlet spigot of the one or more TRP trays. Each of the one

or more flexible hoses has a first quick connect fitting and each of the one or more flexible hoses has a second quick connect fitting. The thermal runaway ducting system further may include a sensor manifold in fluid communication with the one or more flexible hoses and a battery pack exit vent. The sensor manifold has one or more inlet ports in fluid communication with one of the one or more flexible hoses. The sensor manifold has a sensor port configured to allow measurement of a fluid within the sensor manifold using a sensor. The sensor manifold has an outlet port in fluid communication with the battery pack exit vent.

[0022] In another aspect of the present disclosure, the one or more flexible hoses further includes a first flexible hose. The first quick connect fitting of the first flexible hose is in fluid communication with a first outlet spigot of a first TRP tray. The second quick connect fitting of the first flexible hose is in fluid communication with the sensor manifold. The one or more flexible hoses further may include a second flexible hose. The first quick connect fitting of the second flexible hose is in fluid communication with a second outlet of a second TRP tray. The second quick connect fitting of the second flexible hose is in fluid communication with the sensor manifold.

[0023] In another aspect of the present disclosure, the sensor manifold further includes a first inlet port in fluid communication with the second quick connect fitting of the first flexible hose. The sensor manifold further may include a second inlet port in fluid communication with the second quick connect fitting of the second flexible hose.

[0024] Further areas of applicability will become apparent from the description provided herein. It should be understood that the description and specific examples are intended for purposes of illustration only and are not intended to limit the scope of the present disclosure.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0025] The drawings described herein are for illustration purposes only and are not intended to limit the scope of the present disclosure in any way.

[0026] FIG. 1A is a bottom view of a thermal runaway protection system, according to an exemplary embodiment;

[0027] FIG. 1B is a top view of a thermal runaway protection system, according to an exemplary embodiment;

[0028] FIG. 2A is a magnified view of a portion of the thermal runaway protection system of FIG. 1A showing a first exemplary embodiment of a thermal runaway ducting system, according to an exemplary embodiment;

[0029] FIG. 2B is a perspective magnified view of a portion of the top view of the thermal runaway protection system of FIG. 1B is shown, according to an exemplary embodiment;

[0030] FIG. 2C is a magnified view of a portion of the thermal runaway protection system of FIG. 1A showing a second exemplary embodiment of a thermal runaway ducting system, according to an exemplary embodiment;

[0031] FIG. 2D is a magnified view of a portion of the thermal runaway protection system of FIG. 1A showing a third exemplary embodiment of a thermal runaway ducting system, according to an exemplary embodiment;

[0032] FIG. 3 is a bottom perspective view of the thermal runaway protection system of FIG. 1A including a fourth exemplary embodiment of the thermal runaway ducting system, according to an exemplary embodiment;

[0033] FIG. 4A is a side view of a portion of the bottom perspective view of the thermal runaway protection system of FIG. 3 including a first outlet spigot, according to an exemplary embodiment;

[0034] FIG. 4B is a perspective view of a portion of the thermal runaway protection system shown in FIG. 3 including the sensor manifold, according to an exemplary embodiment;

[0035] FIG. 4C is a section (as indicated in FIG. 3) of a first TRP tray, according to an exemplary embodiment;

[0036] FIG. 5 is a bottom perspective view of a fifth exemplary embodiment of the thermal runaway ducting system, according to an exemplary embodiment; and

[0037] FIG. 6 is a schematic diagram of an exemplary vehicle shown with a battery pack including the thermal runaway protection system, according to an exemplary embodiment.

DETAILED DESCRIPTION

[0038] The following description is merely exemplary in nature and is not intended to limit the present disclosure, application, or uses.

[0039] Referring to FIGS. 1A-1B, a bottom view of a thermal runaway protection system **12** is shown in FIG. 1A, and a top view of the thermal runaway protection system **12** is shown in FIG. 1B. The thermal runaway protection system **12** includes a first TRP tray **14a**, a second TRP tray **14b**, and a battery pack casting **16**. The first and second TRP trays **14a**, **14b** are used to capture and direct fluid expelled from one or more electric battery cells (not shown) during a thermal runaway event. In the scope of the present disclosure, fluid expelled during a thermal runaway event may include liquids and/or gases (e.g., carbon dioxide, carbon monoxide, methane, hydrogen, ethylene, and/or the like). The fluid further may include solid particles (e.g., pieces of internal components of the one or more electric battery cells) suspended in the fluid. In an exemplary embodiment, the first and second TRP trays **14a**, **14b** include a tray portion **18** and a lid portion **20**. The tray portion **18** and the lid portion **20** are affixed to each other to form a fluid collection cavity **21** (FIG. 4C), as will be discussed in greater detail below in reference to FIG. 4C. In an exemplary embodiment, the lid portion **20** includes one or more mica plates **22** (FIG. 1B). Each of the one or more electric battery cells (not shown) is disposed above one of the one or more mica plates **22**. During a thermal runaway event, a pressure relief valve on one or more of the electric battery cells (not shown) releases, venting fluid out of the electric battery cell. The force of the venting fluid bursts the mica plate **22** disposed under the electric battery cell, allowing the fluid to vent into the fluid collection cavity **21** (FIG. 4C) of the TRP tray **14**. It is advantageous to quickly and reliably expel fluid from the thermal runaway protection system **12** to mitigate the buildup of high-temperature fluid during a thermal runaway event. Accordingly, the present disclosure provides multiple embodiments of new and improved thermal runaway ducting systems and apparatuses for battery packs.

[0040] Referring to FIG. 2A and with continued reference to FIG. 1A, a magnified view of a portion **24** of the thermal runaway protection system **12** showing a first exemplary embodiment of a thermal runaway ducting system **30** is shown. The first exemplary embodiment of the thermal runaway ducting system **30** includes a first flexible hose **32a**, a second flexible hose **32b**, and a sensor manifold **34**. Each of the flexible hoses **32a**, **32b** has a first quick connect fitting **36a**, a second quick connect fitting **36b**, and a flexible tube **38** in fluid communication with the first quick connect fitting **36a** and the second quick connect fitting **36b**.

[0041] In a non-limiting example, the first and second quick connect fittings **36a**, **36b** include one or more sealing members (e.g., O-rings, gaskets, and/or the like) and provide for a fluid-tight connection without screwing, welding, brazing, and/or the like. In a non-limiting example, the first and second quick connect fittings **36a**, **36b** are configured to provide verification of a successful connection, using, for example, a change in shape, size, position, orientation, or color of one or more components of the first and second quick connect fittings **36a**, **36b**. For example, SELFY Quick Connectors manufactured and/or distributed by ARAYMOND MOBILITY may be used as the first and second quick connect fittings **36a**, **36b**. In a non-limiting example, the flexible tube **38** is made from a thermoplastic of UL94 V1 or better rating. In another non-limiting example, the flexible tube **38** is made from metal. In a non-limiting example, an inner diameter of the flexible tube **38** is between 0.25 inches and 3 inches, inclusive.

[0042] In the first exemplary embodiment of the thermal runaway ducting system **30**, the first quick connect fitting **36a** of the first flexible hose **32a** is in fluid communication with the first TRP tray

14a. In the first exemplary embodiment of the thermal runaway ducting system **30**, the first TRP tray **14a** includes a first outlet spigot **40**. The first quick connect fitting **36a** is removably connected with the first outlet spigot **40** of the first TRP tray **14a**, establishing fluid communication with the first TRP tray **14a** and allowing fluid in the first TRP tray **14a** to be evacuated into the first flexible hose **32a**. In a non-limiting example, the first outlet spigot **40** includes a quick connect fitting configured to engage with the first quick connect fitting **36a**. The second quick connect fitting **36b** of the first flexible hose **32a** is in fluid communication with the sensor manifold **34**, as will be discussed in greater detail below.

[0043] The first quick connect fitting **36a** of the second flexible hose **32b** is in fluid communication with the second TRP tray **14b**. In the first exemplary embodiment of the thermal runaway ducting system **30**, the second TRP tray **14b** includes a first outlet spigot **40**. The first quick connect fitting **36a** of the second flexible hose **32b** is removably connected with the first outlet spigot **40** of the second TRP tray **14b**, establishing fluid communication with the second TRP tray **14b** and allowing fluid in the second TRP tray **14b** to be evacuated into the second flexible hose **32b**. In a non-limiting example, the first outlet spigot **40** includes a quick connect fitting configured to engage with the first quick connect fitting **36a**. The second quick connect fitting **36b** of the second flexible hose **32b** is in fluid communication with the sensor manifold **34**, as will be discussed in greater detail below.

[0044] Referring to FIG. 2B, a perspective magnified view of a portion **42** of the top view of the thermal runaway protection system **12** of FIG. 1B is shown. With reference to FIG. 2B and continued reference to FIG. 2A, the sensor manifold **34** is used to receive fluid from the first and second TRP trays **14a**, **14b**. The sensor manifold **34** includes a first inlet port **44a**, a second inlet port **44b**, a sensor port **46**, and an outlet port **48**. The first and second inlet ports **44a**, **44b** are in fluid communication with first and second flexible hoses **32a**, **32b**. In the first exemplary embodiment of the thermal runaway ducting system **30**, the first inlet port **44a** is in fluid communication with the second quick connect fitting **36b** of the first flexible hose **32a** and the second inlet port **44b** is in fluid communication with the second quick connect fitting **36b** of the second flexible hose **32b**. In a non-limiting example, the first and second inlet ports **44a**, **44b** include quick connect fittings configured to connect with the second quick connect fitting **36b** of the first and second flexible hoses **32a**, **32b**.

[0045] The sensor port **46** (FIG. 2B) is configured to allow measurement of the fluid within the sensor manifold **34** using a sensor **50**. In an exemplary embodiment, the sensor port **46** includes one or more sealing members (e.g., O-rings, gaskets, and/or the like) providing a fluid-tight seal between the sensor **50** and the sensor manifold **34**. In an exemplary embodiment, the sensor **50** is used by a controller (not shown) to measure characteristics of a fluid within the sensor manifold **34**. In a non-limiting example, the sensor **50** includes one or more of: a fluid type sensor, a fluid temperature sensor, a fluid flow rate sensor, and/or the like.

[0046] The outlet port **48** is used to expel fluid from the sensor manifold **34**. The outlet port **48** is in fluid communication with a battery pack exit vent **52** (FIG. 2B) integral with the battery pack casting **16**. In an exemplary embodiment, the outlet port **48** is in fluid communication with a sensor manifold exit spigot **54** (FIG. 2B) using a quick connect fitting **49**. The sensor manifold exit spigot **54** is in fluid communication with the battery pack exit vent **52** using a press-in-place seal. In the scope of the present disclosure, a press-in-place seal includes a compressible material (e.g., rubber or silicone) disposed on a first mating surface (e.g., the sensor manifold exit spigot **54**) configured to fit into a corresponding groove or channel on a second mating surface (e.g., the battery pack exit vent **52**). In a non-limiting example, the press-in-place seal is installed by pressing the first mating surface against the second mating surface (e.g., using fasteners such as screws, bolts, rivets, or the like), such that the compressible material conforms to the shape of the first mating surface and the groove/channel of the second mating surface, forming a fluid-tight seal.

[0047] Referring to FIG. 2C and with continued reference to FIG. 1A, a magnified view of a

portion **24** of the bottom view of the thermal runaway protection system **12** showing a second exemplary embodiment of the thermal runaway ducting system **60** is shown. The second exemplary embodiment of the thermal runaway ducting system **60** includes a first flexible hose **32a'**, a second flexible hose **32b'**, and a sensor manifold **34'**. The description of the first and second flexible hoses **32a**, **32b** provided above in reference to the first exemplary embodiment of the thermal runaway ducting system **30** also applies to the first and second flexible hoses **32a'**, **32b'** of the second exemplary embodiment of the thermal runaway ducting system **60**.

[0048] In the second exemplary embodiment of the thermal runaway ducting system **60**, the first quick connect fitting **36a'** of the first flexible hose **32a'** is in fluid communication with the first TRP tray **14a**. In the second exemplary embodiment of the thermal runaway ducting system **60**, the first TRP tray **14a** includes a first outlet spigot **40'**. The first quick connect fitting **36a'** of the first flexible hose **32a'** is removably connected with the first outlet spigot **40'** of the first TRP tray **14a**, establishing fluid communication with the first TRP tray **14a** and allowing fluid in the first TRP tray **14a** to be evacuated into the first flexible hose **32a'**. The description of the first outlet spigot **40** provided above in reference to the first exemplary embodiment of the thermal runaway ducting system **30** also applies to the first outlet spigot **40'** of the second exemplary embodiment of the thermal runaway ducting system **60**. In a non-limiting example, the first outlet spigot **40'** includes a quick connect fitting configured to engage with the first quick connect fitting **36a'** of the first flexible hose **32a'**. In the second exemplary embodiment of the thermal runaway ducting system **60**, the second TRP tray **14b** includes a second outlet spigot **62**. In a non-limiting example, the second outlet spigot **62** includes two quick connect fittings. The second quick connect fitting **36b'** of the first flexible hose **32a'** is in fluid communication with the second outlet spigot **62** of the second TRP tray **14b**.

[0049] The first quick connect fitting **36a'** of the second flexible hose **32b'** is in fluid communication with the second TRP tray **14b** using the second outlet spigot **62** of the second TRP tray **14b**, allowing fluid in the first TRP tray **14a** and the second TRP tray **14b** to be evacuated into the second flexible hose **32b'**. In a non-limiting example, the second outlet spigot **62** includes two quick connect fittings configured to engage with the second quick connect fitting **36b'** of the first flexible hose **32a'** and the first quick connect fitting **36a'** of the second flexible hose **32b'**. The second quick connect fitting **36b'** of the second flexible hose **32b'** is in fluid communication with the sensor manifold **34'**, as will be discussed in greater detail below.

[0050] It should be understood that the second outlet spigot **62** of the second TRP tray **14b** differs from the first outlet spigot **40'** in that the second outlet spigot **62** allows fluid communication between the first flexible hose **32a'**, the second flexible hose **32b'**, and the second TRP tray **14b**. Therefore, in the second exemplary embodiment of the thermal runaway ducting system **60**, fluid evacuated from the first TRP tray **14a** may enter the second TRP tray **14b** at the second outlet spigot **62**.

[0051] The sensor manifold **34'** of the second exemplary embodiment of the thermal runaway ducting system **60** includes a first inlet port **44a'**, the sensor port **46**, and the outlet port **48** as described above in reference to the first exemplary embodiment of the thermal runaway ducting system **30**. In the second exemplary embodiment of the thermal runaway ducting system **60**, the first inlet port **44a'** is in fluid communication with the second quick connect fitting **36b'** of the second flexible hose **32b'**. The description of the first inlet port **44a** provided above in reference to the first exemplary embodiment of the thermal runaway ducting system **30** also applies to the first inlet port **44a'** of the second exemplary embodiment of the thermal runaway ducting system **60**. The description of the sensor port **46** and the outlet port **48** provided above in reference to the sensor manifold **34** of the first exemplary embodiment of the thermal runaway ducting system **30** is also applicable to the sensor manifold **34'** of the second exemplary embodiment of the thermal runaway ducting system **60**.

[0052] Referring to FIG. 2D and with continued reference to FIG. 1A, a magnified view of the

portion **24** of the thermal runaway protection system **12** showing a third exemplary embodiment of the thermal runaway ducting system **70** is shown. The third exemplary embodiment of the thermal runaway ducting system **70** includes a first flexible hose **32a''**, a tee-joint hose **32c''**, and a sensor manifold **34''**. The description of the first flexible hose **32a** provided above in reference to the first exemplary embodiment of the thermal runaway ducting system **30**, also applies to the first flexible hose **32a''** of the third exemplary embodiment of the thermal runaway ducting system **70**. The tee-joint hose **32c''** includes a first quick connect fitting **72a**, a second quick connect fitting **72b**, a third quick connect fitting **72c**, and a flexible tube **74**. The first quick connect fitting **72a**, second quick connect fitting **72b**, and third quick connect fitting **72c** are in fluid communication with each other via the flexible tube **74**. In a non-limiting example, the flexible tube **74** is made from a thermoplastic of UL94 V1 or better rating. In another non-limiting example, the flexible tube **74** is made from metal. In a non-limiting example, an inner diameter of the flexible tube **74** is between 0.25 inches and 3 inches, inclusive.

[0053] In the third exemplary embodiment of the thermal runaway ducting system **70**, the first quick connect fitting **36a''** of the first flexible hose **32a''** is in fluid communication the first quick connect fitting **72a** of the tee-joint hose **32c''**, allowing fluid in the tee-joint hose **32c''** to be evacuated into the first flexible hose **32a''**. The second quick connect fitting **72b** of the tee-joint hose **32c''** is in fluid communication with the first TRP tray **14a**. In the third exemplary embodiment of the thermal runaway ducting system **70**, the first TRP tray **14a** includes a first outlet spigot **40''**. The second quick connect fitting **72b** of the tee-joint hose **32c''** is removably connected with the first outlet spigot **40''** of the first TRP tray **14a**. The description of the first outlet spigot **40** provided above in reference to the first exemplary embodiment of the thermal runaway ducting system **30** also applies to the first outlet spigot **40''** of the third exemplary embodiment of the thermal runaway ducting system **70**. In the third exemplary embodiment of the thermal runaway ducting system **70**, the second TRP tray **14b** includes a first outlet spigot **40''**. The third quick connect fitting **72c** of the tee-joint hose **32c''** is removably connected with the first outlet spigot **40''** of the second TRP tray **14b**. Therefore, in the third exemplary embodiment of the thermal runaway ducting system **70**, fluid evacuated from the first TRP tray **14a** may enter the second TRP tray **14b** at the tee-joint hose **32c''**. Furthermore, in the third exemplary embodiment of the thermal runaway ducting system **70**, fluid evacuated from the second TRP tray **14b** may enter the first TRP tray **14a** at the tee-joint hose **32c''**.

[0054] The sensor manifold **34''** of the third exemplary embodiment of the thermal runaway ducting system **70** includes a first inlet port **44a''**, the sensor port **46**, and the outlet port **48** as described above in reference to the second exemplary embodiment of the thermal runaway ducting system **60**. The description of the sensor manifold **34'** provided above in reference to the second exemplary embodiment of the thermal runaway ducting system **60** is also applicable to the sensor manifold **34''** of the third exemplary embodiment of the thermal runaway ducting system **70**. The first inlet port **44a''** of the sensor manifold **34''** is in fluid communication with the second quick connect fitting **36b''** of the first flexible hose **32a''**.

[0055] Referring to FIG. 3, a bottom perspective view of the thermal runaway protection system **12** including a fourth exemplary embodiment of the thermal runaway ducting system **80** is shown. The fourth exemplary embodiment of the thermal runaway ducting system **80** includes a first flexible hose **32a'''**, a second flexible hose **32b'''**, and a sensor manifold **34'''**. The description of the first and second flexible hoses **32a**, **32b** provided above in reference to the first exemplary embodiment of the thermal runaway ducting system **30** also applies to the first and second flexible hoses **32a'''**, **32b'''** of the fourth exemplary embodiment of the thermal runaway ducting system **80**.

[0056] In the fourth exemplary embodiment of the thermal runaway ducting system **80**, the first quick connect fitting **36a'''** of the first flexible hose **32a'''** is in fluid communication with the first TRP tray **14a**. In the fourth exemplary embodiment of the thermal runaway ducting system **80**, the first TRP tray **14a** includes a first outlet spigot **40'''**. The first quick connect fitting **36a'''** of the first

flexible hose **32a'''** is removably connected with the first outlet spigot **40'''** of the first TRP tray **14a**, allowing fluid in the first TRP tray **14a** to be evacuated into the first flexible hose **32a'''**. The second quick connect fitting **36b'''** of the first flexible hose **32a'''** is in fluid communication with the sensor manifold **34''**, as will be discussed in greater detail below. The first quick connect fitting **36a'''** of the second flexible hose **32b'''** is in fluid communication with the second TRP tray **14b**. In the fourth exemplary embodiment of the thermal runaway ducting system **80**, the second TRP tray **14b** includes a first outlet spigot **40'''**. The first quick connect fitting **36a'''** of the second flexible hose **32b'''** is removably connected with the first outlet spigot **40'''** of the second TRP tray **14b**, allowing fluid in the second TRP tray **14b** to be evacuated into the second flexible hose **32b'''**. The second quick connect fitting **36b'''** of the second flexible hose **32b'''** is in fluid communication with the sensor manifold **34'''**, as will be discussed in greater detail below.

[0057] Referring to FIG. 4A, a side view of a portion **82** of the bottom perspective view of the thermal runaway protection system **12** of FIG. 3 including the first outlet spigot **40'''** is shown. The first outlet spigot **40'''** has a first portion **84a** and a second portion **84b**. The first portion **84a** is in fluid communication with the second portion **84b** with a hollow internal cavity. The first portion **84a** is disposed within the fluid collection cavity **21** formed between a convex surface **86b** of the tray portion **18** of the first and second TRP trays **14a**, **14b** and an interior lid surface **88** of the lid portion **20** of the first and second TRP trays **14a**, **14b**. In an exemplary embodiment, the first portion **84a** of the first outlet spigot **40'''** includes a solid particulate filter **90** for preventing solid particles from entering the outlet spigot **40'''**. In a non-limiting example, the solid particulate filter **90** is a sieve-like structure including one or more holes allowing fluid to pass through. Each of the one or more holes are configured with an area less than or equal to a predetermined maximum particle cross-sectional area. The predetermined maximum particle cross-sectional area defines a cross-sectional area of a largest particle which is permitted to enter the first outlet spigot **40'''**. In another exemplary embodiment, the first portion **84a** of the first outlet spigot **40'''** further includes one or more angled particle deflection surfaces (not shown) disposed proximally to the solid particulate filter **90** for deflecting solid particles away from the solid particulate filter **90**.

[0058] In an exemplary embodiment, the second portion **84b** of the first outlet spigot **40'''** includes a quick connect fitting **92** configured to engage with the first quick connect fitting **36a'''**. In a non-limiting example, as shown in FIG. 4A, the quick connect fitting **92** is substantially perpendicular to the first and second TRP trays **14a**, **14b**. In the scope of the present disclosure, substantially perpendicular means that an angle **94** measured between the quick connect fitting **92** and the first and second TRP trays **14a**, **14b** is approximately ninety degrees within a predetermined tolerance (e.g., $\pm 10\%$).

[0059] Referring again to FIG. 3, in the fourth exemplary embodiment of the thermal runaway ducting system **80**, the first outlet spigot **40'''** is disposed proximally to an outside edge of each of the first and second TRP trays **14a**, **14b**, allowing for improved ergonomic access for connection of the first quick connect fitting **36a'''** of the first flexible hose **32a'''** and the first quick connect fitting **36a'''** of the second flexible hose **32b'''** during assembly of the thermal runaway protection system **12**.

[0060] Referring to FIG. 4B, a perspective view of a portion **96** of the thermal runaway protection system **12** shown in FIG. 3 including the sensor manifold **34'''** is shown. The sensor manifold **34'''** of the fourth exemplary embodiment of the thermal runaway ducting system **80** includes a first inlet port **44a'''**, a second inlet port **44b'''**, a sensor port **46'''**, and an outlet port **48'''**. In the fourth exemplary embodiment of the thermal runaway ducting system **80**, the first inlet port **44a'''** is in fluid communication with the second quick connect fitting **36b'''** of the first flexible hose **32a'''**. The second inlet port **44b'''** is in fluid communication with the second quick connect fitting **36b'''** of the second flexible hose **32b'''**. In a non-limiting example, the first and second inlet ports **44a'''**, **44b'''** include quick connect fittings configured to connect with the second quick connect fitting **36b'''** of the first and second flexible hoses **32a'''**, **32b'''**.

[0061] The sensor port **46'''** is configured to allow measurement of the fluid within the sensor manifold **34'''** using the sensor **50**, as discussed above. In an exemplary embodiment, the sensor port **46'''** includes one or more sealing members (e.g., O-rings, gaskets, and/or the like) providing a fluid-tight seal between the sensor **50** and the sensor manifold **34'''**. In an exemplary embodiment, the sensor **50** is used by a controller (not shown) to measure characteristics of a fluid within the sensor manifold **34'''**. In a non-limiting example, the sensor **50** includes one or more of: a fluid type sensor, a fluid temperature sensor, a fluid flow rate sensor, and/or the like.

[0062] The outlet port **48'''** is used to expel fluid from the sensor manifold **34'''**. In the fourth exemplary embodiment of the thermal runaway ducting system **80**, the outlet port **48'''** is in fluid communication with a battery pack exit vent **52'''** integral with the battery pack casting **16**. In an exemplary embodiment, the outlet port **48'''** is in fluid communication with the battery pack exit vent **52'''** using a press-in-place seal between the sensor manifold **34'''** and the battery pack casting **16**. In the scope of the present disclosure, a press-in-place seal includes a compressible material (e.g., rubber or silicone) disposed on a first mating surface (e.g., the sensor manifold **34'''**) configured to fit into a corresponding groove or channel on a second mating surface (e.g., the battery pack casting **16**). In a non-limiting example, the press-in-place seal is installed by pressing the first mating surface against the second mating surface (e.g., using fasteners such as screws, bolts, rivets, or the like), such that the compressible material conforms to the shape of the first mating surface and the groove/channel of the second mating surface, forming a fluid-tight seal.

[0063] Referring to FIG. 4C, a section (as indicated in FIG. 3) of the first TRP tray **14a** is shown. In an exemplary embodiment, the first and second TRP trays **14a**, **14b** further include one or more filtering fin members **110** disposed on the interior lid surface **88** of the lid portion **20** and extending into the fluid collection cavity **21**. The one or more filtering fin members **110** function to disrupt a flow of solid particles within the fluid collection cavity **21** in order to prevent the solid particles from entering the outlet spigot **40'''**. In an exemplary embodiment, the first and second TRP trays **14a**, **14b** include a plurality of filtering fin members **110** distributed throughout the first and second TRP trays **14a**, **14b**. In another exemplary embodiment, the first and second TRP trays **14a**, **14b** include a plurality of filtering fin members **110** disposed proximally to the outlet spigot **40'''**. In a non-limiting example, the plurality of filtering fin members **110** are disposed proximally to the outlet spigot **40'''** in a radial or “U-shaped” pattern to induce a swirling motion in the fluid. The swirling motion causes solid particles to impact the plurality of filtering fin members **110**, exiting the fluid flow.

[0064] In a non-limiting example, the one or more filtering fin members **110** are made of a same material as the lid portion **20** (e.g., plastic, polymer, metal, composite, and/or the like). It should be understood that the one or more filtering fin members **110** may be used with any of the embodiments of the thermal runaway ducting system discussed in the present disclosure.

[0065] Referring to FIG. 5, a bottom perspective view of a fifth exemplary embodiment of the thermal runaway ducting system **120** is shown. The fifth exemplary embodiment of the thermal runaway ducting system **120** includes a first flexible hose **32a'''**, a second flexible hose (not shown) and a sensor manifold **34'''**. The description of the first flexible hose **32a'''** provided above in reference to the first exemplary embodiment of the thermal runaway ducting system **30** also applies to the first flexible hose **32a'''** of the fifth exemplary embodiment of the thermal runaway ducting system **120**.

[0066] In the fifth exemplary embodiment of the thermal runaway ducting system **120**, the first quick connect fitting **36a'''** of the first flexible hose **32a'''** is in fluid communication with the first TRP tray **14a**. In the fifth exemplary embodiment of the thermal runaway ducting system **120**, the first TRP tray **14a** includes a first outlet spigot **40'''**. The first quick connect fitting **36a'''** of the first flexible hose **32a'''** is removably connected to the first outlet spigot **40'''** of the first TRP tray **14a**, allowing fluid in the first TRP tray **14a** to be evacuated into the first flexible hose **32a'''**. The description of the first outlet spigot **40'''** provided above in reference to the fourth exemplary

embodiment of the thermal runaway ducting system **80** also applies to the first outlet spigot **40'''** of the fifth exemplary embodiment of the thermal runaway ducting system **120**. The second quick connect fitting **36b'''** of the first flexible hose **32a'''** is in fluid communication with the sensor manifold **34'''**, as will be discussed in greater detail below. In the fifth exemplary embodiment of the thermal runaway ducting system **120**, the second TRP tray **14b** (not shown in FIG. 5) includes a first outlet spigot (not shown). The first quick connect fitting (not shown) of the second flexible hose (not shown) is removably connected to the first outlet spigot (not shown) of the second TRP tray **14b**, allowing fluid in the second TRP tray **14b** to be evacuated into the second flexible hose (not shown). The second quick connect fitting (not shown) of the second flexible hose (not shown) is in fluid communication with the sensor manifold **34'''**, as will be discussed in greater detail below.

[0067] In the fifth exemplary embodiment of the thermal runaway ducting system **120**, the sensor manifold **34'''** is integral with a battery pack casting **16'''**. In the scope of the present disclosure, integral with means that multiple parts are combined as a single structure. The battery pack casting **16'''** includes a first inlet port **44a'''** and a second inlet port (not shown) for connection with the first flexible hose **32a'''** and the second flexible hose (not shown). The battery pack casting **16'''** further includes a hollow cavity **122** for containing fluids evacuated from the first flexible hose **32a'''** and the second flexible hose (not shown). The battery pack casting **16'''** further includes a sensor port **46'''** for receiving the sensor **50**. The battery pack casting **16'''** further includes a battery pack exit vent **52'''** for expelling fluids from the thermal runaway protection system **12**. The second quick connect fitting **36b'''** of the first flexible hose **32a'''** is in fluid communication with the first inlet port **44a'''**. The second quick connect fitting (not shown) of the second flexible hose (not shown) is in fluid communication with the second inlet port (not shown). It should be understood that embodiments where the sensor manifold **34'''** is integral with the battery pack casting **16'''**, as discussed in reference to the fifth exemplary embodiment of the thermal runaway ducting system **120**, may be used with any of the embodiments of the thermal runaway ducting system discussed in the present disclosure.

[0068] Referring to FIG. 6, schematic diagram of an exemplary vehicle **200** is shown with a battery pack **202** including the thermal runaway protection system **12**. While a passenger vehicle is illustrated, it should be appreciated that the vehicle **200** may be any type of vehicle without departing from the scope of the present disclosure. In an exemplary embodiment, the battery pack **202** including the thermal runaway protection system **12** is disposed on an underside of the vehicle **200**.

[0069] The embodiments of the thermal runaway ducting system discussed in the present disclosure offer several advantages. Use of the quick connect fittings allows for verifiable connection of the components. Furthermore, the ducting system may be pressure and/or leak tested after assembly to ensure fluid-tight construction. The embodiments of the thermal runaway ducting system discussed in the present disclosure are robust to intrusion of potting material which may be used within the thermal runaway protection system **12**, ensuring that fluid venting paths remain unobstructed. Additionally, the use of the solid particulate filter and/or the one or more filtering fin members prevents entry of solid particles into the fluid venting paths. Furthermore, the flexible hoses provide for ease of assembly and mitigate clearance issues due to manufacturing variation. Additionally, orientation of the quick connect fittings substantially perpendicular to the one or more TRP trays enables line of sight for robotic assembly of the thermal runaway protection system **12**.

[0070] The description of the present disclosure is merely exemplary in nature and variations that do not depart from the gist of the present disclosure are intended to be within the scope of the present disclosure. Such variations are not to be regarded as a departure from the spirit and scope of the present disclosure.

Claims

1. A thermal runaway ducting system for a battery pack having one or more thermal runaway protection (TRP) trays, the thermal runaway ducting system comprising: one or more flexible hoses in fluid communication with the one or more TRP trays; and a sensor manifold in fluid communication with the one or more flexible hoses and a battery pack exit vent.
2. The thermal runaway ducting system of claim 1, each of the one or more flexible hoses further comprising: a first quick connect fitting; a second quick connect fitting; and a flexible tube in fluid communication with the first quick connect fitting and the second quick connect fitting, wherein the flexible tube provides fluid communication between the first quick connect fitting and the second quick connect fitting.
3. The thermal runaway ducting system of claim 2, the one or more flexible hoses further comprising: a first flexible hose, wherein the first quick connect fitting of the first flexible hose is in fluid communication with a first TRP tray, and wherein the second quick connect fitting of the first flexible hose is in fluid communication with the sensor manifold; and a second flexible hose, wherein the first quick connect fitting of the second flexible hose is in fluid communication with a second TRP tray, and wherein the second quick connect fitting of the second flexible hose is in fluid communication with the sensor manifold.
4. The thermal runaway ducting system of claim 3, the sensor manifold further comprising: a first inlet port in fluid communication with the second quick connect fitting of the first flexible hose; and a second inlet port in fluid communication with the second quick connect fitting of the second flexible hose.
5. The thermal runaway ducting system of claim 2, the one or more flexible hoses further comprising: a first flexible hose, wherein the first quick connect fitting of the first flexible hose is in fluid communication with a first TRP tray, and wherein the second quick connect fitting of the first flexible hose is in fluid communication with a second TRP tray; and a second flexible hose, wherein the first quick connect fitting of the second flexible hose is in fluid communication with the second TRP tray, and wherein the second quick connect fitting of the second flexible hose is in fluid communication with the sensor manifold.
6. The thermal runaway ducting system of claim 5, the sensor manifold further comprising: a first inlet port in fluid communication with the second quick connect fitting of the second flexible hose.
7. The thermal runaway ducting system of claim 2, the one or more flexible hoses further comprising: a tee-joint hose in fluid communication with a first TRP tray and a second TRP tray; and a first flexible hose, wherein the first quick connect fitting of the first flexible hose is in fluid communication with the tee-joint hose, and wherein the second quick connect fitting of the first flexible hose is in fluid communication with the sensor manifold.
8. The thermal runaway ducting system of claim 7, the sensor manifold further comprising: a first inlet port in fluid communication with the second quick connect fitting of the first flexible hose.
9. The thermal runaway ducting system of claim 1, the sensor manifold further comprising: one or more inlet ports in fluid communication with one of the one or more flexible hoses; a sensor port configured to allow measurement of a fluid within the sensor manifold using a sensor; and an outlet port in fluid communication with the battery pack exit vent.
10. The thermal runaway ducting system of claim 9, the thermal runaway ducting system further comprising: a battery pack casting, wherein the battery pack exit vent is integral with the battery pack casting, and wherein the outlet port of the sensor manifold is in fluid communication with the battery pack exit vent using a press-in-place seal.
11. The thermal runaway ducting system of claim 9, the thermal runaway ducting system further comprising: a battery pack casting, wherein the battery pack exit vent is integral with the battery pack casting, and wherein the sensor manifold is integral with the battery pack casting.

12. A thermal runaway ducting system for a battery pack, the thermal runaway ducting system comprising: one or more thermal runaway protection (TRP) trays; one or more flexible hoses in fluid communication with the one or more TRP trays using quick connect fittings; and a sensor manifold in fluid communication with the one or more flexible hoses and a battery pack exit vent using quick connect fittings.

13. The thermal runaway ducting system of claim 12, each of the one or more TRP trays further comprising: an outlet spigot, wherein the outlet spigot has a first portion disposed within one of the one or more TRP trays and a second portion disposed outside of one of the one or more TRP trays, and wherein the second portion is in fluid communication with the one or more flexible hoses.

14. The thermal runaway ducting system of claim 13, the first portion of the outlet spigot further comprising: a solid particulate filter for preventing solid particles from entering the outlet spigot.

15. The thermal runaway ducting system of claim 13, the second portion of the outlet spigot further comprising: a quick connect fitting configured to engage with one of the one or more flexible hoses.

16. The thermal runaway ducting system of claim 15, wherein the quick connect fitting is substantially perpendicular to one of the one or more TRP trays.

17. The thermal runaway ducting system of claim 13, each of the one or more TRP trays further comprising: a tray portion having a convex surface; a lid portion having an interior lid surface, wherein the lid portion is affixed to the tray portion such that a fluid collection cavity is formed between the interior lid surface of the lid portion and the convex surface of the tray portion; and one or more filtering fin members disposed on the interior lid surface of the lid portion, wherein the one or more filtering fin members are configured to prevent solid particles from entering the outlet spigot.

18. A thermal runaway ducting system for a battery pack, the thermal runaway ducting system comprising: one or more thermal runaway protection (TRP) trays, wherein each of the one or more TRP trays has an outlet spigot, wherein the outlet spigot has a first portion disposed within one of the one or more TRP trays and a second portion disposed outside of one of the one or more TRP trays, and wherein the second portion of the outlet spigot has a quick connect fitting; one or more flexible hoses in fluid communication with the outlet spigot of the one or more TRP trays, wherein each of the one or more flexible hoses has a first quick connect fitting, and wherein each of the one or more flexible hoses has a second quick connect fitting; and a sensor manifold in fluid communication with the one or more flexible hoses and a battery pack exit vent, wherein the sensor manifold has one or more inlet ports in fluid communication with one of the one or more flexible hoses, wherein the sensor manifold has a sensor port configured to allow measurement of a fluid within the sensor manifold using a sensor, and wherein the sensor manifold has an outlet port in fluid communication with the battery pack exit vent.

19. The thermal runaway ducting system of claim 18, the one or more flexible hoses further comprising: a first flexible hose, wherein the first quick connect fitting of the first flexible hose is in fluid communication with a first outlet spigot of a first TRP tray, and wherein the second quick connect fitting of the first flexible hose is in fluid communication with the sensor manifold; and a second flexible hose, wherein the first quick connect fitting of the second flexible hose is in fluid communication with a second outlet of a second TRP tray, and wherein the second quick connect fitting of the second flexible hose is in fluid communication with the sensor manifold.

20. The thermal runaway ducting system of claim 19, the sensor manifold further comprising: a first inlet port in fluid communication with the second quick connect fitting of the first flexible hose; and a second inlet port in fluid communication with the second quick connect fitting of the second flexible hose.
